



AirFly Insights: Data Visualization and Analysis of Airline Operations

Infosys Springboard | Virtual Internship 6.0 | Batch 13

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01 | Project Overview

About the Project

- Analyzed airline dataset to understand flight performance, delays, cancellations, and seasonal trends
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- Developed interactive dashboards using Python, Power BI, and Streamlit for clear visualization
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- Identified key insights on busiest routes, airport congestion, and operational patterns to support data-driven decisions

Duration: 8 Weeks | Start: 02 Feb 2026 | Submission: 30 Mar 2026



Data Analytics

Clean & transform World Bank datasets



Dashboard Development

Interactive Power BI analytical dashboard



Web Application

Full-stack React + Node.js web portal



Secure Admin Portal

JWT auth, user management & telemetry

02 | Project Workflow

Start

Data Collection

Data Preprocessing

Data Structuring & Transformation

Power BI Dashboard Development

Testing & Review

Documentation & Final Report

End



03 | Master Sheet Preparation

Data Preparation Steps

- 1. Source —** World Bank database – Gini Index, GDP per capita, Unemployment rate, Income share by decile
- 2. Cleaning —** Removed irrelevant columns, standardized field names, converted inconsistent formats
- 3. Missing Values —** Applied linear interpolation for sparse years; used country averages where interpolation not feasible
- 4. Long Format —** Converted all indicator files from wide to long format (Country × Year rows)
- 5. Merge —** Joined all datasets on Country + Year into a single unified master sheet
- 6. Validation —** Verified numeric types, confirmed Year as integer, checked for blank/null values in Power BI

Key Columns in Master Sheet

Column Name	Description
Airport Name	Alrport name
Pilot Name	Assigned pilot
Route	Origin to destination path
IsDelayed	Delay indicator (0/1)
IsCancelled	Cancellation indicator (0/1)
Flight Status	On-time / delayed / cancelled
Country Name	Country of airport
Arrival Airport	Destination airport
Day of Departure	Day of week
Passenger ID	Unique passenger identifier
Age	Passenger age

05 | Dashboard Overview & Global Insights

Cancellation Distribution : Donut Chart
Monthly Delay Trend : Line Chart
Flight Status Distribution by Month : Stacked Bar Chart
Flights Distribution by Day : Pie Chart
Average Delay by Airport : Bar Chart

Indicators Comparison Section
Flight Delay vs Cancellation Comparison : Bar Chart
Flight Status Distribution Across Days : Pie Chart
Airport Traffic vs Average Delay Analysis : Bar Chart
Monthly Flight Activity vs Delay Trend : Line Chart

Global Insights

Most flights are on time, but delays are common

Flight activity shows seasonal trends

Some routes and airports are highly busy

Delays are more frequent than cancellations



06 | Key Features & Highlights

- Real-time filtering by Country and Month for dynamic insights
- KPI metrics: Total Flights, Delayed Flights, Cancelled Flights
- Visualization of trends using pie charts, line charts, bar charts, and heatmaps
- Identification of busiest routes and high-traffic airports
- Clear analysis of delay patterns and seasonal trends
- User-friendly and visually appealing dashboard design
- Interactive dashboard for analyzing flight performance, delays, and cancellations

07 | Challenges Faced & Solutions Implemented

CHALLENGES



Data Inconsistency & Missing Values Analysis



Feature Understanding & Structuring Indicators



Dashboard Design Complexity



Visualization Clarity



Insight Extraction

SOLUTIONS



Cleaned dataset and handled missing values for accurate analysis



Created meaningful features like Month, Route, and Delay indicators



Organized visuals into structured sections with clear layout and filters



Selected appropriate charts (pie, bar, line, heatmap) for better interpretation



Focused on key patterns like delays, traffic, and seasonal trends for meaningful conclusions

08 | Project Outcomes

4

Milestones
Completed

8

Weeks
Internship Duration

11

DAX Measures

18

Columns
Master Sheet Fields



Data Analytics Outcome

- Cleaned and prepared dataset
- Created useful features for analysis
- Identified key patterns and trends



Dashboard Outcome

- Developed interactive Power BI dashboard
- Visualized key metrics and trends
- Enabled easy data exploration with filters



Web Application Outcome

- Built interactive dashboard using Streamlit
- Enabled real-time data filtering and visualization
- Provided user-friendly interface for analysis



Learning Outcome

- Learned data cleaning and analysis using Python
- Gained experience in data visualization tools
- Improved dashboard design and storytelling skills

10 | Conclusion

Project Summary

- This project successfully analyzed airline operations using data visualization techniques. It identified key patterns in delays, cancellations, and traffic trends. The project followed an end-to-end approach from data cleaning to dashboard development using Python, Power BI, and Streamlit. The final outcome is an interactive and user-friendly dashboard that provides clear insights for better decision-making.



Future Potential

- Add real-time flight data
- Use machine learning for delay prediction
- Enhance dashboard with advanced features



Infosys Springboard Virtual Internship 6.0 — Completed Successfully

Batch 13 | March 2026

Thank You!

For Your Attention